EXP 5: Comparative Analysis of Different Types of Prompting Patterns

NAME:D.SANJUTHA

REFERENCE NO:212225240136

Aim

To test and compare how different prompt patterns respond to various tasks by analyzing the responses generated by ChatGPT using Naïve prompts (broad or unclear prompts) and Basic prompts (clear, detailed, structured prompts). The responses are evaluated based on quality, accuracy, and depth.

AI TOO REQUIRED

CHAT GPT

1. Prompt Types

Naïve Prompt

A naïve prompt is a **simple or vague instruction** that gives very little context or details to the AI model.

Example:
"Explain photosynthesis."

These prompts may lead to **short or general answers** because the model has limited instructions.

Basic Prompt

A basic prompt is **clear, structured, and detailed**. It provides instructions and context to guide the AI model to generate a better response.

Example:
"Explain the process of photosynthesis in simple terms suitable for a school student and include the main steps involved."

These prompts usually produce **more accurate, detailed, and organized responses**.

Test Scenarios

Scenario 1 – Creative Story Generation

Naïve Prompt

Write a story about a robot.

Output (ChatGPT):
A robot named Robo lived in a small town. Unlike other robots, Robo was curious about humans and wanted to learn emotions. One day, Robo helped a lost child find their parents. From that moment, people realized robots could be kind and helpful too.

Basic Prompt

Write a short creative story about a friendly robot who helps people in a city. The story should include a problem, solution, and a happy ending.

Output (ChatGPT):
In a busy city lived a robot named Nova who loved helping people. One day, the city traffic system stopped working, causing confusion everywhere. Nova used its advanced sensors to guide cars and pedestrians safely through the streets. Soon engineers repaired the system, and everyone thanked Nova for saving the day. From then on, Nova became the city's most trusted helper.

Scenario 2 – Answering a Factual Question

Naïve Prompt

What is photosynthesis?

Output:
Photosynthesis is the process by which plants make their food using sunlight, carbon dioxide, and water.

Basic Prompt

Explain the process of photosynthesis, including the materials required, the role of sunlight, and the products formed.

Output:
Photosynthesis is the process by which green plants produce their own food using sunlight. It occurs mainly in the leaves inside chloroplasts. Plants use sunlight as energy to convert carbon dioxide from the air and water from the soil into glucose (food). Oxygen is released as a byproduct during this process.

Scenario 3 – Summarizing a Concept

Naïve Prompt

Explain Artificial Intelligence.

Output:
Artificial Intelligence (AI) is a technology that allows machines to perform tasks that normally require human intelligence, such as learning, reasoning, and decision making.

Basic Prompt

Provide a short summary of Artificial Intelligence including its definition, common applications, and benefits.

Output:
Artificial Intelligence is the simulation of human intelligence in machines that can learn, analyze data, and make decisions. AI is used in many applications such as voice assistants, medical diagnosis, self-driving cars, and recommendation systems. It helps improve efficiency, automate tasks, and support better decision-making.

Scenario 4 – Advice / Recommendation

Naïve Prompt

How to study better?

Output:
You can study better by staying focused, practicing regularly, and avoiding distractions.

Basic Prompt

Give five practical tips for students to improve their study habits and manage time effectively.

Output:

1. Create a daily study schedule.
2. Study in a quiet and distraction-free environment.
3. Take short breaks to maintain concentration.
4. Revise lessons regularly instead of last-minute studying.
5. Practice problems and test yourself to improve understanding.